Chapter 2

First-order Logic

Propositional logic is not powerful enough to capture all inferences possible from a given set of premises. The following is an example.

All dogs are animals.
Tuffy is a dog.
Therefore, Tuffy is an animal.

The above argument is intuitively valid. However, if we denote

p: All dogs are animals.
q: Tuffy is a dog.
r: Tuffy is an animal.

Then, *r cannot* be shown to be a logical consequence of *p* and *q* within the framework of propositional logic. This is because the above sentences are formed from simpler constituents which encompass concepts such as properties of objects and relationships between objects, and the validity of the above logical inference cannot be established without breaking the sentences into such simpler constituents. Propositional Logic is too primitive to handle that.

In this chapter, we shall consider the ***First-order Logic*** (abbreviated as ***FOL***). First-order Logic extends Propositional Logic by including logic components which allow one to speak of objects, properties of objects, and relationship between objects. It will become obvious that much of our daily and mathematical language can be symbolized by first-order logic.

2.1 Predicates

Predicates are used to specify properties of objects or relationships between objects, which frequently are the basic constituents of a sentence.

Definition: A ***predicate*** is a function[2] whose value is *true* or *false*.

2 A formal definition of function will be given in Part II. In this chapter, the definition of function in high school algebra will be used.

Example 2.1.1 The following are predicates. Recall that the variables x, y, z are called ***arguments*** of the predicates (functions) in high school algebra.

$$Dog(x) = \begin{cases} true & if\ x\ is\ a\ dog \\ false & otherwise \end{cases}$$

$$Animal(x) = \begin{cases} true & if\ x\ is\ an\ animal \\ false & otherwise \end{cases}$$

$$Succ(x,y) = \begin{cases} true & if\ y = x+1 \\ false & otherwise \end{cases}$$

$$Greater(x,y) = \begin{cases} true & if\ x > y \\ false & if\ x \leq y \end{cases}$$

$$Equal(x,y) = \begin{cases} true & if\ x = y \\ false & if\ x \neq y \end{cases}$$

$$Sum(x,y,z) = \begin{cases} true & if\ x+y = z \\ false & if\ x+y \neq z \end{cases}$$

For convenience, we shall use a shorter notation to define predicates. Let $P(x_1, x_2, \cdots, x_n)$ be an n-arguments predicate, and '$condition_T$' ('$condition_F$', respectively) represents the condition for which $P(x_1, x_2, \cdots, x_n)$ assumes the value *true* (*false*, respectively). Then, instead of writing

$$P(x_1, x_2, \cdots, x_n) = \begin{cases} true & if\ condition_T \\ false & if\ condition_F \end{cases}$$

we shall write $P(x_1, x_2, \cdots, x_n)\ :\ condition_T$.

For example, the predicates *Dog* and *Greater* can be defined as follow:

$$Dog(x)\ :\ x\ is\ a\ dog;$$
$$Greater(x,y)\ :\ x\ >\ y.$$

Informally speaking, a predicate is a sentence which contains one or more variables, and will become a proposition when each of its variables is replaced by a specific element from its domain. A predicate without an argument can also be regarded as a proposition. The truth values, *true* (T) and *false* (F), can be regarded as predicates which map every element in the domain to the value *true* and *false*, respectively.

Example 2.1.2

(i) The domain of the predicate *Succ* is the set of positive integers. If in *Succ*(*x*,*y*), x is replaced by 3, and y is replaced by 4, then *Succ*(3,4) is the proposition 'the successor of 3 is 4' which is *true*.

(ii) The domain of the predicate *Sum* is the set of integers. If in *Sum*(*x*,*y*,*z*), x is replaced by -1, y is replaced by 5, and z is replaced by 6, then *Sum*(-1,5,6) is the proposition '-1 plus 5 is 6' which is *false*.

(iii) The 0-argument predicate (a predicate without argument)

$$Prop = \begin{cases} true & if\ 1+1 = 2 \\ false & otherwise \end{cases}$$

can be regarded as the proposition "1+1=2."

As the values of predicates are true or false, predicates can be used as operands of the logical operators to form more complicated constructs. e.g. Using the predicates *Greater* and *Sum*, we can form $((Sum(x, y, z) \wedge Greater(y, 0)) \Rightarrow Greater(z, x))$ which asserts that "if $x + y = z$ and $y > 0$, then $z > x$."

2.2 Quantifiers

Quantifiers are used to express concepts, such as 'every', 'all', 'some'.

There are two quantifiers: the ***universal quantifier*** and the ***existential quantifier***. The universal quantifier is denoted by $\forall$ which is pronounced as '*for all*' or '*for every*' or '*for each.*' The existential quantifier is denoted by $\exists$ which is pronounced as '*for some*' or '*there exists*' or '*there is at least one.*'

The quantifiers are used in the following ways.

$$(\forall x)F \qquad (\exists x)F$$

where x is a variable, and F is a First-order-logic well-formed formulas (to be defined below). As with wffs in Propositional Logic, First-order-logic well-formd formulas are evaluated to the *true* or *false* value.

The construct $(\forall x)F$ asserts that for every x, F is evaluated to the *true* value.

The construct $(\exists x)F$ asserts that there exists an x for which F is evaluated to the *true* value.

2.3 First-order Logic Well-formed Formulas

Using predicates, the quantifiers and the logical operators, we can construct expressions in First-order logic. Expressions in First-order logic are called ***First-order logic well-formed formulas*** (abbreviated as ***FOL-wffs***) and are defined as follows.

Definition:

(a) The predicates (with variables or constants substituted for their arguments) and the truth values T(*true*) and F(*false*) are FOL-wffs.

(b) If α and β are FOL-wffs, then $\sim \alpha$, $(\alpha \vee \beta)$, $(\alpha \wedge \beta)$, $(\alpha \Rightarrow \beta)$, and $(\alpha \Leftrightarrow \beta)$ are FOL-wffs.

(c) If F is a FOL-wff, then $(\forall x)F$ and $(\exists x)F$ are FOL-wffs.

(d) No construct other than those that can be defined by means of a finite number of applications of (a), (b) and (c) is a FOL-wff.

Example 2.3.3 The following are FOL-wffs. The predicates appeared in the FOL-wffs are defined in Example 2.1.1

(i) $(Dog(x) \Rightarrow Animal(x))$

(ii) $(\forall x)(\exists y)(Succ(x, y) \wedge (\forall z)(Succ(x, z) \Rightarrow Equal(y, z)))$

(iii) $\sim (\exists x)Succ(x, 0)$

(iv) $(\forall x)(\sim Equal(x, 0) \Rightarrow (\exists y)(Succ(y, x) \wedge (\forall z)(Succ(z, x) \Rightarrow Equal(y, z))))$

(v) $(((\forall x)(Dog(x) \Rightarrow Animal(x)) \wedge Dog(Tuffy)) \Rightarrow Animal(Tuffy))$

(vi) $(\exists x)Greater(y, z)$

Note that in case (c) of the definition of FOL-wff, it is not required that F must contain the variable x, nor is it required to contain only the variable x. Therefore, the FOL-wff in Example 2.3.3(vi) is valid, even though $Greater(y, z)$ does not contain x, and contains the variables y and z.

To reduce the amount of parentheses involved in writing FOL-wffs, we adopt the following operator precedence.

Operator	Precedence
$\forall\ \exists$	1
$\sim$	2
$\wedge$	3
$\vee$	4
$\Rightarrow$	5
$\Leftrightarrow$	6

As with Propositional Logic, the binary operators are left-associative, while the unary operator and the quantifiers are right-associative.

Moreover, $(\forall w)(\forall x)\cdots(\forall z)F$ and $(\exists w)(\exists x)\cdots(\exists z)F$ may, respectively, be abbreviated as $(\forall w, x, \cdots, z)F$ and $(\exists w, x, \cdots, z)F$, or $\forall w, x, \cdots, z, F$ and $\exists w, x, \cdots, z, F$.

Definition: In $(\forall x)F$ ($(\exists x)F$, respectively), the quantifier $\forall$ ($\exists$, respectively) is said to ***quantify*** the variable x. The ***scope*** of $(\forall x)$ ($(\exists x)$, respectively) is the FOL-wff F. In a FOL-wff, an occurrence of a variable x is ***free*** if and only if it is not within the scope of a quantifier quantifying x and is not quantified by a quantifier. An occurrence of a variable x is ***bound*** iff it is not free. A variable is *free* in a FOL-wff if it has a free occurrence in the FOL-wff. A variable is *bound* in a FOL-wff if it has a bound occurrence in the FOL-wff.

Example 2.3.4 Let P, Q, R and S be predicates.

(i) In $(\forall x)(P(x) \Rightarrow S(x))$, the scope of the quantifier $(\forall x)$ is $(P(x) \Rightarrow S(x))$. The x in $(\forall x)$ and the two x's in $(P(x) \Rightarrow S(x))$ are all bound.

(ii) In $(\forall x)(\exists y)(P(x, y) \wedge R(x, y) \Rightarrow (\exists z)(Q(x, z) \vee \sim Q(y, z)))$. The scope of $(\forall x)$ is $(\exists y)(P(x, y) \wedge R(x, y) \Rightarrow (\exists z)(Q(x, z) \vee \sim Q(y, z)))$. The scope of $(\exists y)$ is $(P(x, y) \wedge R(x, y) \Rightarrow (\exists z)(Q(x, z) \vee \sim Q(y, z)))$. the scope of $(\exists z)$ is $(Q(x, z) \vee \sim Q(y, z))$. All the occurrences of x, y, z are bound.

(iii) In $(\forall x)P(x, y)$, the scope of $(\forall x)$ is $P(x, y)$. The two occurrences of x are bound but the y is free.

(iv) In $(\forall x)P(x, y) \wedge (\exists y)Q(y)$, the scope of $(\forall x)$ is $P(x, y)$ and the scope of $(\exists y)$ is $Q(y)$. All the x's are bound. The first occurrence of y is free but the second and third occurrences of y are bound.

2.4 Evaluation of FOL-wffs

To evaluate a FOL-wff, we have to first give each predicate and constant in the wff a meaning. To be more specific, we must first define: (i) the ***Universe of discourse***, denoted by U, which is the set from which the variables and constants in the wff draw their values; (ii) all the constants, and all the predicates in the wff. Moreover, the FOL-wff must not contain any free variable. For instance, we cannot evaluate $Greater(0, a)$ without assigning the constant a an integer value, and we cannot evaluate $(\forall x)(\exists y)(P(x, y) \Rightarrow \sim Q(y))$ without the definitions of predicates P and Q. Such an assignment of meaning to the constants and predicates in a FOL-wff is called an interpretation of the wff. Formally,

Definition: An ***interpretation*** I of a FOL-wff consists of

(i) a definition for the universe of discourse, U;
(ii) an assignment of a value from U to each constant in the wff;
(iii) a definition for each predicate in the wff;

Let ω be a FOL-wff. Given an interpretation I of ω, the truth value of ω can be determined recursively, using the following rules:

(a) if α, β are sub-wffs of ω whose truth values have been determined, then the truth values of the following sub-wffs of ω (if any) can be determined by using the definitions of the logical operators given in Chapter 1.

$$\sim \alpha,\ (\alpha \vee \beta),\ (\alpha \wedge \beta),\ (\alpha \Rightarrow \beta),\text{ and } (\alpha \Leftrightarrow \beta)$$

(b) Let F be a FOL-wff and $F(a)$ be the FOL-wff resulting from F after every *free* occurrence of x in it is replaced by an a.

i. The sub-wff $(\forall x)F$ is evaluated to *true* if and only if $F(a)$ is evaluated to *true* for *every* a in U.
ii. The sub-wff $(\exists x)F$ is evaluated to *true* if and only if $F(a)$ is evaluated to *true* for *at least one* a in U.

Remark: If F *does not* contain any *free* occurrence of x, then the truth value of F is independent of x. This means that either F evaluates to *true* for every x in U, or F evaluates to *false* for every x in U. As a result, the truth value of $(\forall x)F$ is identical to that of F. Consequently, $(\forall x)F$ evaluates to *true* iff F evaluates to *true*. Similarly, $(\exists x)F$ evaluates to *true* iff F evaluates to *true*.

Example 2.4.5

(i) Evaluate $(\forall x)(\exists y)P(x, y)$ using the following interpretation.

The Universe of discourse: $\mathsf{U} = \{a, b\}$

Definition of predicate P:

x	y	$P(x,y)$
a	a	T
a	b	F
b	a	T
b	b	T

(a) Let $x = a$. We obtain $(\exists y)P(a, y)$.
Let $y = a$. Then $P(a, a) \equiv \mathsf{T}$.

$\therefore$ $(\exists y)P(a, y)$ is evaluated to *true*.

(b) Let $x = b$. We obtain $(\exists y)P(b, y)$.
Let $y = b$. Then $P(b, b) \equiv \mathsf{T}$.
$\therefore$ $(\exists y)P(b, y)$ is evaluated to *true*.

Since $(\exists y)P(x, y)$ is evaluated to *true* for every x in U, $(\forall x)(\exists y)P(x, y)$ is evaluated to T. □

(ii) Evaluate $(\forall x)(P(x) \Rightarrow (\forall y)Q(x, y))$ using the following interpretation.
The Universe of discourse: U = $\{a, b\}$

Definition of predicate P:

x	$P(x)$
a	T
b	F

Definition of predicate Q:

x	y	$Q(x,y)$
a	a	T
a	b	T
b	a	F
b	b	T

(a) Let $x = a$. We obtain $(P(a) \Rightarrow (\forall y)Q(a, y))$. Since $P(a) \equiv \mathsf{T}$, we have to determine the value of $(\forall y)Q(a, y)$ in order to determine the truth value of $(P(a) \Rightarrow (\forall y)Q(a, y))$.

i. Let $y = a$. Then $Q(a, y) \equiv Q(a, a) \equiv \mathsf{T}$.
ii. Let $y = b$. Then $Q(a, y) \equiv Q(a, b) \equiv \mathsf{T}$.

Hence, $(\forall y)Q(a, y)$ is evaluated to *true*.
Consequently, $(P(a) \Rightarrow (\forall y)Q(a, y))$ is evaluated to *true*.

(b) Let $x = b$. We obtain $(P(b) \Rightarrow (\forall y)Q(b, y))$. Since $P(b) \equiv \mathsf{F}$, $(P(b) \Rightarrow (\forall y)Q(b, y))$ is evaluated to *true*, regardless of the value of $(\forall y)Q(b, y)$.

Since $(P(x) \Rightarrow (\forall y)Q(x, y))$ is evaluated to true for every x in U, $(\forall x)(P(x) \Rightarrow (\forall y)Q(x, y))$ is thus evaluated to *true*. □

(iii) Evaluate $(\forall x)(\forall y)(P(x, y) \Rightarrow (\exists z)(Q(x, y) \wedge Q(y, z)))$ using the following interpretation.
The Universe of discourse: U = $\{a, b\}$

Definition of predicate P:

x	y	$P(x,y)$
a	a	T
a	b	F
b	a	T
b	b	T

Definition of predicate Q:

x	y	$Q(x,y)$
a	a	T
a	b	F
b	a	T
b	b	F

(a) Let $x = a$ in $(\forall x)(\forall y)(P(x,y) \Rightarrow (\exists z)(Q(x,y) \wedge Q(y,z)))$,
we obtain $(\forall y)(P(a,y) \Rightarrow (\exists z)(Q(a,y) \wedge Q(y,z)))$.

i. Let $y = a$ in $(\forall y)(P(a,y) \Rightarrow (\exists z)(Q(a,y) \wedge Q(y,z)))$.
We obtain $(P(a,a) \Rightarrow (\exists z)(Q(a,a) \wedge Q(a,z)))$.
Since $P(a,a) \equiv \mathsf{T}$, we have to determine the value of $(\exists z)(Q(a,a) \wedge Q(a,z))$.
Let $z = a$, we have $(Q(a,a) \wedge Q(a,z)) \equiv (Q(a,a) \wedge Q(a,a)) \equiv \mathsf{T} \wedge \mathsf{T} \equiv \mathsf{T}$.
Therefore, $(\exists z)(Q(a,a) \wedge Q(a,z)) \equiv \mathsf{T}$.
Hence, $(P(a,a) \Rightarrow (\exists z)(Q(a,a) \wedge Q(a,z)))$ is evaluated to *true*.

ii. Let $y = b$ in $(\forall y)(P(a,y) \Rightarrow (\exists z)(Q(a,y) \wedge Q(y,z)))$.
We obtain $(P(a,b) \Rightarrow (\exists z)(Q(a,b) \wedge Q(b,z)))$.
Since $P(a,b) \equiv \mathsf{F}$, $(P(a,b) \Rightarrow (\exists z)(Q(a,b) \wedge Q(b,z)))$ is evaluated to *true*.

Hence, $(P(a,y) \Rightarrow (\exists z)(Q(a,y) \wedge Q(y,z)))$ is evaluated to *true* for every y in U.
$(\forall y)(P(a,y) \Rightarrow (\exists z)(Q(a,y) \wedge Q(y,z)))$ is thus evaluated to *true*.

(b) Let $x = b$ in $(\forall x)(\forall y)(P(x,y) \Rightarrow (\exists z)(Q(x,y) \wedge Q(y,z)))$.
We obtain $(\forall y)(P(b,y) \Rightarrow (\exists z)(Q(b,y) \wedge Q(y,z)))$.

i. Let $y = a$ in $(\forall y)(P(b,y) \Rightarrow (\exists z)(Q(b,y) \wedge Q(y,z)))$.
We obtain $(P(b,a) \Rightarrow (\exists z)(Q(b,a) \wedge Q(a,z)))$.
Since $P(b,a) \equiv \mathsf{T}$, we have to determine the value of $(\exists z)(Q(b,a) \wedge Q(a,z))$.
Let $z = a$. We obtain $(Q(b,a) \wedge Q(a,z)) \equiv (Q(b,a) \wedge Q(a,a)) \equiv \mathsf{T} \wedge \mathsf{T} \equiv \mathsf{T}$.
Therefore, $(\exists z)(Q(b,a) \wedge Q(a,z)) \equiv \mathsf{T}$.
Hence, $(P(b,a) \Rightarrow (\exists z)(Q(b,a) \wedge Q(a,z)))$ is evaluated to *true*.

ii. Let $y = b$ in $(\forall y)(P(b,y) \Rightarrow (\exists z)(Q(b,y) \wedge Q(y,z)))$,
we obtain $(P(b,b) \Rightarrow (\exists z)(Q(b,b) \wedge Q(b,z)))$.
Since $P(b,b) \equiv \mathsf{T}$, we have to determine the value of $(\exists z)(Q(b,b) \wedge Q(b,z))$.
Let $z = a$. We obtain $Q(b,b) \wedge Q(b,a) \equiv \mathrm{F} \wedge \mathrm{T} \equiv \mathrm{F}$.
Let $z = b$. We obtain $Q(b,b) \wedge Q(b,b) \equiv \mathrm{F} \wedge \mathrm{F} \equiv \mathrm{F}$.
Therefore, $(\exists z)(Q(b,b) \wedge Q(b,z)) \equiv \mathsf{F}$.
Hence, $(P(b,b) \Rightarrow (\exists z)(Q(b,b) \wedge Q(b,z)))$ is evaluated to *false*.

Since $(P(b,y) \Rightarrow (\exists z)(Q(b,y) \wedge Q(y,z)))$ is evaluated to *false* when $y = b$, $(\forall y)(P(b,y) \Rightarrow (\exists z)(Q(b,y) \wedge Q(y,z)))$ is thus evaluated to *false*. This implies that $(\forall y)(P(x,y) \Rightarrow (\exists z)(Q(x,y) \wedge Q(y,z)))$ is evaluated to *false* when $x = b$. Hence, $(\forall x)(\forall y)(P(x,y) \Rightarrow (\exists z)(Q(x,y) \wedge Q(y,z)))$ is evaluated to *false*. □

Remark: Actually, Case (b)ii, by itself, is sufficient to determine the value of $(\forall x)(\forall y)(P(x,y) \Rightarrow (\exists z)(Q(x,y) \wedge Q(y,z)))$. We can thus shorten the solution by presenting only the evaluation in Case (b)ii. This is particularly important when the universe of discourse U contains an infinite number of elements. This method of presenting only *one* element from the universe of discourse to show that a FOL-wff of the form $(\forall x)\alpha(x)$ is evaluated to *false* is called ***proof by counterexample***.

(iv) Evaluate $(\exists x)(\forall y)(Q(x,y) \vee P(x,y))$ using the interpretation provided in (iii).
Let $x = b$.
As $Q(b,a) \vee P(b,a) \equiv \mathrm{T} \vee \mathrm{T} \equiv \mathrm{T}$ and $Q(b,b) \vee P(b,b) \equiv \mathrm{F} \vee \mathrm{T} \equiv \mathrm{T}$,

$(\forall y)(Q(x,y) \vee P(x,y))$ is evaluated to *true* when $x = b$.

Therefore, $(\exists x)(\forall y)(Q(x,y) \vee P(x,y))$ is evaluated to *true*.

Remark: In the above example, we presented just *one* value of x to show that $(\exists x)(\forall y)(Q(x,y) \vee P(x,y))$ is evaluated to *true*. This method of presenting just *one* element from the universe of discourse to show that a FOL-wff of the form $(\exists x)\alpha(x)$ is evaluated to *true* is called ***proof by example***.

In each of the above four examples, we evaluated FOL-wffs of the form $(\forall x)\alpha(x)$ by examining every possible way of assigning values from the universe of discourse to the variables in the wffs. Although we did not present the solution in a tabulated form, it is essentially a tabulation method similar to the truth table method. Clearly, such a tabulation method can handle only a finite number of cases. If the universe of discourse U contains an infinite number of elements, the tabulation method will become infeasible (not just impractical). The following are two examples.

(v) Evaluate $(\forall x)(\exists y)P(x,y)$ where $P(x,y) : x < y$ and U is the set of all integers. (Note that P represents the "*less than*" relation among integers.)

Since U contains an infinite number of integers, we cannot evaluate $(\forall x)(\exists y)P(x,y)$ by evaluating $(\exists y)P(x,y)$ for every integer x. Otherwise, we would have to perform an infinite number of evaluations.

(vi) Evaluate $(\forall x)(\exists y)P(x,y)$ where $P(x,y) : x > y$ and U is the set of all positive integers. (Note that P represents the "*greater than*" relation among positive integers.)

Again, as U contains an infinite number of integers, we cannot evaluate $(\forall x)(\exists y)P(x,y)$ by evaluating $(\exists y)P(x,y)$ for every integer x.

Fortunately, based on the properties of positive integers, we discover that $(\forall x)(\exists y)P(x,y)$ is evaluated to *false*. We can thus use the *proof by counterexample* method. The solution is as follows.

Since 1 is a positive integer and no positive integer y is less than 1, therefore, there does not exist a y in U such that $1 > y$. In other words, $(\exists y)P(1,y)$ is evaluated to *false*. This implies that, when $x = 1$, $(\exists y)P(x,y)$ is evaluated to *false*. Hence, $(\forall x)(\exists y)P(x,y)$ is evaluated to *false*. ■

2.5 Satisfiable, Valid and Inconsistent FOL-wffs

Definition: A ***closed FOL-wff*** is a FOL-wff without free variable.

Let F be a FOL-wff and I be an interpretation of F. If F is a closed FOL-wff, then as was shown in the above examples, F is evaluated to a truth value under I. However, if F is not closed, then F cannot be evaluated under I unless each of its free variables is assigned a value from the Universe of discourse of I. Clearly, it is quite likely that for some assignments of values to the free variables, F is evaluated to *true*, while for others, F is evaluated to *false*.

Definition: Let F be a FOL-wff and I be an interpretation of F. F evaluates to *true* if for *every* assignment of values to the free variables in F, the resulting closed wff evaluates to *true*.

Therefore, in determining the truth value of a FOL-wff with free variables, the free variables are treated as if they were quantified by the universal quantifier.

Definition: A FOL-wff F is ***satisfiable*** iff there exists an interpretation I such that F evaluates to true under I. I is called a ***model*** of F and I ***satisfies*** F.

Definition: A FOL-wff F is ***vaild*** iff every interpretation of F satisfies F. A FOL-wff F is ***inconsistent*** (***unsatisfiable***, a ***contradiction***) iff no interpretation of F satisfies F.

Example 2.5.6

(i) Example 2.4.5(i) shows that $(\forall x)(\exists y)P(x, y)$ is satisfiable.

(ii) Example 2.4.5(ii) shows that $(\forall x)(P(x) \Rightarrow (\forall y)Q(x, y))$ is satisfiable.

(iii) Example 2.4.5(iii) shows that $(\forall x)(\forall y)(P(x, y) \Rightarrow (\exists z)(Q(x, y) \wedge Q(y, z)))$ is *not* a valid FOL-wff.

(iv) Example 2.4.5(v) shows that $(\forall x)(\exists y)P(x, y)$ is satisfiable, and Example 2.4.5(vi) shows that it is *not* a valid FOL-wff.

(v) Let α, β be FOL-wffs. Prove that $(\alpha \vee \beta) \Leftrightarrow (\beta \vee \alpha)$ is valid.

Let I be any interpretation for $(\alpha \vee \beta) \Leftrightarrow (\beta \vee \alpha)$. Since each of α and β evaluates to *true* or *false* under I, there are altogether 4 possible outcomes for the values of α and β. We can thus construct a table to determine the value of $(\alpha \vee \beta) \Leftrightarrow (\beta \vee \alpha)$ for each of the four possible outcomes. This table turns out to be identical to the truth table used in evaluating the wff $(\alpha \vee \beta) \Leftrightarrow (\beta \vee \alpha)$ in propositional logic, as is evident below.

α	β	$(\alpha \vee \beta)$	$(\beta \vee \alpha)$	$(\alpha \vee \beta) \Leftrightarrow (\beta \vee \alpha)$
F	F	F	F	T
F	T	T	T	T
T	F	T	T	T
T	T	T	T	T

The right-most column in the above table clearly shows that $(\alpha \vee \beta) \Leftrightarrow (\beta \vee \alpha)$ evaluates to *true* for every possible values of α and β.

Therefore, $(\alpha \vee \beta) \Leftrightarrow (\beta \vee \alpha)$ is evaluated to *true* for every interpretation. It is thus a valid FOL-wff. ■

2.6 First-order Logic Equivalences and Inference Rules

Definition: Two FOL-wffs α and β are ***(logically) equivalent***, denoted by $\alpha \equiv \beta$, iff $(\alpha \Leftrightarrow \beta)$ is valid.

A list of FOL equivalences is given below.

First-order Logic Equivalences

Remark: In the following equivalences, $\alpha(x)$ represents a FOL-wff which may contain a free occurrence of x, and $\alpha(y)$ is the FOL-wff resulting from $\alpha(x)$ after every free occurrence of x in it is replaced by a y. Note that $\alpha(x)$ may contain other free variables.

E1 — E20 of propositional logic

FE1. $(\forall x)\alpha(x) \equiv (\forall y)\alpha(y)$ ($\alpha(x)$ does not contain y)
FE2. $(\exists x)\alpha(x) \equiv (\exists y)\alpha(y)$ ($\alpha(x)$ does not contain y)
FE3. $(\forall x)\alpha(x) \vee \beta \equiv (\forall x)(\alpha(x) \vee \beta)$ (β contains no free occurrence of x)
FE4. $(\exists x)\alpha(x) \vee \beta \equiv (\exists x)(\alpha(x) \vee \beta)$ (β contains no free occurrence of x)
FE5. $(\forall x)\alpha(x) \wedge \beta \equiv (\forall x)(\alpha(x) \wedge \beta)$ (β contains no free occurrence of x)
FE6. $(\exists x)\alpha(x) \wedge \beta \equiv (\exists x)(\alpha(x) \wedge \beta)$ (β contains no free occurrence of x)
FE7. $\sim (\forall x)\alpha(x) \equiv (\exists x) \sim \alpha(x)$
FE8. $\sim (\exists x)\alpha(x) \equiv (\forall x) \sim \alpha(x)$
FE9. $(\forall x)\alpha(x) \wedge (\forall x)\beta(x) \equiv (\forall x)(\alpha(x) \wedge \beta(x))$
FE10. $(\exists x)\alpha(x) \vee (\exists x)\beta(x) \equiv (\exists x)(\alpha(x) \vee \beta(x))$

Proof of E1 — E20: To prove that each of the equivalences, E1–E20, remains as an equivalence in FOL, we can use the truth table. This was illustarated in Example 2.5.6(v) in which E10 is shown to be an equivalence in FOL.

Proof of FE1: We are to prove that $(\forall x)\alpha(x) \Leftrightarrow (\forall y)\alpha(y)$ is valid.

Since $\alpha(x)$ contains no occurrence of y, an occurrence of x is free in $\alpha(x)$ iff the occurrence of y that replaces it is free in $\alpha(y)$. Moreover, every free occurrence of y in $\alpha(y)$ corresponds to a free occurrence of x in $\alpha(x)$. As a result, let I be an interpretation and U be its universal of discourse. Then, for every t in U, the wff obtained from $\alpha(x)$ by replacing every free occurrence of x with t, denoted by $\alpha(t)$, is identical to that obtained from $\alpha(y)$ by replacing every free occurrence of y with t.

Hence, $(\forall x)\alpha(x)$ is evaluated to *true*
iff $\alpha(t)$ is evaluated to *true* for every t in U
iff $(\forall y)\alpha(y)$ is evaluated to *true*.

Therefore, $(\forall x)\alpha(x) \Leftrightarrow (\forall y)\alpha(y)$ is evaluated to *true* under any interpretation. i.e. $(\forall x)\alpha(x) \equiv (\forall y)\alpha(y)$.

Proof of FE2: Similar to that of FE1.

Proof of FE3: We are to prove that $(\forall x)\alpha(x) \vee \beta \Leftrightarrow (\forall x)(\alpha(x) \vee \beta)$ is valid.

Given any interpretation I. Let U be the universal of discourse in I.

$(\forall x)\alpha(x) \vee \beta$ is evaluated to *true*
iff $(\forall x)\alpha(x)$ is evaluated to *true* or β is evaluated to *true* (Definition of $\vee$)
iff $\alpha(x)$ is evaluated to *true* for every x in U or β is evaluated to *true* (Definition of $(\forall x)F$).

(i) Suppose $\alpha(x)$ is evaluated to *true* for every x in U. Then $\alpha(x) \vee \beta$ is evaluated to *true* for every x in U (Definition of $\vee$).

(ii) Suppose β is evaluated to *true*. Then as β contains no free occurrence of x (i.e the evaluation of β is independent of the value of x), β is evaluated to *true* for every x in U. Therefore, $\alpha(x) \vee \beta$ is evaluated to *true* for every x in U (Definition of $\vee$).

Conversely, suppose $\alpha(x) \vee \beta$ is evaluated to *true* for every x in U.

(i) If β is evaluated to *true* for at least one value of x, say a, then as β contains no free occurrence of x, β is evaluated to *true* for every x in U.

(ii) If β is evaluated to *false* for every x in U, then $\alpha(x)$ must be evaluated to *true* for every x in U (Definition of $\vee$).

It follows that $\alpha(x)$ is evaluated to *true* for every x in U or β is evaluated to *true*
iff $\alpha(x) \vee \beta$ is evaluated to *true* for every x in U
iff $(\forall x)(\alpha(x) \vee \beta)$ is evaluated to *true* (Definition of $(\forall x)F$).

Hence, $(\forall x)\alpha(x) \vee \beta \Leftrightarrow (\forall x)(\alpha(x) \vee \beta)$ is evaluated to *true* under any interpretation.

i.e. $(\forall x)\alpha(x) \vee \beta \equiv (\forall x)(\alpha(x) \vee \beta)$.

Proof of FE4, FE5, FE6: Similar to that of FE3.

Proof of FE7: We are to prove that $\sim (\forall x)\alpha(x) \Leftrightarrow (\exists x) \sim \alpha(x)$ is valid.

Given any interpretation I. Let U be the universal of discourse in I.

$\sim (\forall x)\alpha(x)$ is evaluated to *true*
iff $(\forall x)\alpha(x)$ is evaluated to *false* (Definition of $\sim$)
iff $\alpha(x)$ is evaluated to *false* for at least one x in U (Definition of $(\forall x)F$)
iff $\sim \alpha(x)$ is evaluated to *true* for at least one x in U (Definition of $\sim$)
iff $(\exists x) \sim \alpha(x)$ is evaluated to *true*. (Definition of $(\exists x)F$)

Hence, $\sim (\forall x)\alpha(x) \Leftrightarrow (\exists x) \sim \alpha(x)$ is evaluated to *true* under any interpretation.

i.e. $\sim (\forall x)\alpha(x) \equiv (\exists x) \sim \alpha(x)$.

Proof of FE8: Similar to that of FE7.

Proof of FE9: We are to prove that $(\forall x)\alpha(x) \wedge (\forall x)\beta(x) \Leftrightarrow (\forall x)(\alpha(x) \wedge \beta(x))$ is valid.

Given any interpretation I. Let U be the universal of discourse in I.

$(\forall x)\alpha(x) \wedge (\forall x)\beta(x)$ is evaluated to *true*
iff $(\forall x)\alpha(x)$ is evaluated to *true* and $(\forall x)\beta(x)$ is evaluated to *true* (Definition of $\wedge$)
iff $\alpha(x)$ is evaluated to *true* for every x in U and $\beta(x)$ is evaluated to *true* for every x in U (Definition of $(\forall x)F$)
iff $(\alpha(x) \wedge \beta(x))$ is evaluated to *true* for every x in U (Definition of $\wedge$)
iff $(\forall x)(\alpha(x) \wedge \beta(x))$ is evaluated to *true*. (Definition of $(\forall x)F$)

Hence, $(\forall x)\alpha(x) \wedge (\forall x)\beta(x) \Leftrightarrow (\forall x)(\alpha(x) \wedge \beta(x))$ is evaluated to *true* under any interpretation.

i.e. $(\forall x)\alpha(x) \wedge (\forall x)\beta(x) \equiv (\forall x)(\alpha(x) \wedge \beta(x))$.

Proof of FE10: Similar to that of FE9.

Definition: Let $\alpha(x)$ be a FOL-wff and t be a variable. $\alpha(x)$ is ***free for*** t if no free occurrence of x in $\alpha(x)$ is within the scope of a quantifier quantifying t. In other words, $\alpha(x)$ is free for t if no occurrence of t that replaces a free occurrence of x in $\alpha(x)$ is bound in the resulting FOL-wff.

Example:

(i) Obviously, $\alpha(x)$ is free for x. It is also free for every constant t.

(ii) In $(\forall x)(P(x) \Rightarrow (\exists y)(Q(x) \vee (\exists z)(\sim Q(z) \vee Q(y))))$, the FOL-wff $(P(x) \Rightarrow (\exists y)(Q(x) \vee (\exists z)(\sim Q(z) \vee Q(y))))$ is *not* free for y because the free occurrence of x in $Q(x)$ lies within the scope of $(\exists y)$. However, it is free for z.

Definition: Let $p_1, p_2, \cdots, p_k$ and q be FOL-wffs. We say that $p_1, p_2, \cdots, p_k$ ***logically implies*** q, or q is a ***direct consequence*** of $p_1, p_2, \cdots, p_k$, denote by $\frac{p_1, p_2, \cdots, p_k}{q}$, if $(p_1 \wedge p_2 \wedge \cdots \wedge p_k) \Rightarrow q$ is valid. We call $\frac{p_1, p_2, \cdots, p_k}{q}$ an ***inference rule***.

A list of FOL *inference rules* is listed below.

First-order Logic Inference Rules

I1 — I6 of Propositional logic with p, q, r being FOL-wffs.

Gen: *Generalization*

$$\frac{\alpha(x)}{(\forall x)\alpha(x)} \qquad x \; is \; a \; variable$$

UI: *Universal Instantiation*

$$\frac{(\forall x)\alpha(x)}{\alpha(t)} \qquad \alpha(x) \; is \; free \; for \; t \;\; (t \text{ is a variable or a constant})$$

where $\alpha(t)$ results from $\alpha(x)$ after *all* free occurrences of x in $\alpha(x)$ are replaced by t.

EQ: *Existential Quantification*

$$\frac{\alpha(t)}{(\exists x)\alpha(x)} \qquad \alpha(t) \; is \; free \; for \; x \;\; (t \text{ is a variable or a constant})$$

where $\alpha(x)$ results from $\alpha(t)$ after *some* (may be *all*) free occurrences of t in $\alpha(t)$ are replaced by x.

Proof of I1 — I6: The proofs are similar to those of E1–E20 for First-order Logic.

Proof of Gen: By definition, we are to prove that $\alpha(x) \Rightarrow (\forall x)\alpha(x)$ is valid.

Let I be any interpretation for $\alpha(x) \Rightarrow (\forall x)\alpha(x)$.

Suppose $\alpha(x)$ is evaluated to *true* in I. Then, for each assignment of values to the free variables in $\alpha(x)$, $\alpha(x)$ is evaluated to *true*. (Definition of evaluation of $\alpha(x)$)

This implies that for each assignment of value to the variable x, and for each assignment of values to the other free variables, $\alpha(x)$ is evaluated to *true*,

which implies $(\forall x)\alpha(x)$ is evaluated to *true* for each assignment of values to the free variables in $(\forall x)\alpha(x)$.

Therefore $(\forall x)\alpha(x)$ is evaluated to *true*.

Hence, $\alpha(x) \Rightarrow (\forall x)\alpha(x)$ is evaluated to *true* under I.

Consequently, $\alpha(x) \Rightarrow (\forall x)\alpha(x)$ is valid. ■

Proof of UI: By definition, we are to prove that $(\forall x)\alpha(x) \Rightarrow \alpha(t)$ is valid.

Let I be any interpretation for $(\forall x)\alpha(x) \Rightarrow \alpha(t)$.

Suppose $(\forall x)\alpha(x)$ is evaluated to *true* under I.

Then for any assignment of values to its free variables, $(\forall x)\alpha(x)$ is evaluated to *true*.

It follows that for any assignment of values to the free variables other than x, $\alpha(x)$ is evaluated to *true* for every value of x in U. Now,

(i) If t is a constant in U, then for any assignment of values to the free variables other than x, $\alpha(x)$ is evaluated to *true* when x takes the value t. This implies that $\alpha(t)$ is evaluated to *true*.

(ii) If t is a variable, then as $\alpha(x)$ is free for t, every occurrence of t which replaces a free occurrence of x in $\alpha(x)$ remains free in $\alpha(t)$. This implies that for any assignment of values to the free variables other than x, the value of $\alpha(t)$ is the same as that of $\alpha(x)$ when x and t are given the same value.

Since we have already obtained: for any assignment of values to the other free variables, $\alpha(x)$ is evaluated to *true*. Therefore, for every value of t in U and for any assignment of values to the other free variables, $\alpha(t)$ is evaluated to *true*. This implies that $\alpha(t)$ evaluates to *true*.

Hence, $(\forall x)\alpha(x) \Rightarrow \alpha(t)$ evaluates to *true* for any interpretation. i.e. $(\forall x)\alpha(x) \Rightarrow \alpha(t)$ is valid. ■

Proof of EQ: By definition, we are to prove that $\alpha(t) \Rightarrow (\exists x)\alpha(x)$ is valid.

First, from the proof of UI, we have $(\forall x)\alpha(x) \Rightarrow \alpha(t)$ is a valid FOL-wff. Note that $\alpha(x)$ may contain some occurrences of t.

By substituting $\sim \alpha(x)$ ($\sim \alpha(t)$, respectively) for $\alpha(x)$ ($\alpha(t)$, respectively) in the above FOL-wff, we obtain $(\forall x) \sim \alpha(x) \Rightarrow \sim \alpha(t)$ is a valid FOL-wff.

It follows that $(\forall x) \sim \alpha(x) \Rightarrow \sim \alpha(t)$ is evaluated to *true* in any interpretation I.

By FE8, $\sim (\exists x)\alpha(x) \Rightarrow \sim \alpha(t)$ is evaluated to *true* in any interpretation I.

By E19, $\alpha(t) \Rightarrow (\exists x)\alpha(x)$ is evaluated to *true* in any interpretation I.

As a result, $\alpha(t) \Rightarrow (\exists x)\alpha(x)$ is valid. ■

2.7 Formal Proof

In Example 2.4.5(v), we showed that owing to the fact that there can be infinitely many interpretations for a FOL-wff and for each interpretation, there can be infinitely many elements in the universe of discourse, there is no effective procedure similar to the truth table to determine if a FOL-wff is valid. A purely syntactic approach is thus required. Obviously, to make the syntactic approach useful, it should be sound and *hopefully* complete. In other words, if we can derive a formal proof for a FOL-wff, then the FOL-wff must be valid, and vice versa.

Although the equivalences and inference rules introduced in Section 2.6 are sufficient for the development of a formal proof theory for First-order-logic, we shall include the following useful inference rule. This rule is included for convenience.

EI: *Existential Instantiation*

$$\frac{(\exists x)\alpha(x)}{\alpha(c)} \qquad c \text{ is a constant}$$

where $\alpha(c)$ results from $\alpha(x)$ after *all* free occurrences of x in $\alpha(x)$ are replaced by c.

The inference rule **EI** essentially states that if $(\exists x)\alpha(x)$ is true, then we can always find a constant c for which $\alpha(x)$ is evaluated to true.

The terminology *substitution*, *theorem* and the notations $\Gamma \vdash \alpha$, $\vdash \alpha$ introduced in propositional logic can be used here. The terminology *derivation* (*proof*), however, needs some modifications.

Definition: A ***derivation*** or ***proof*** of a FOL-wff α from a set of FOL-wffs Γ is a sequence of FOL-wffs, $\alpha_1, \alpha_2, \cdots, \alpha_n$ such that

(i) $\alpha_n = \alpha$, and

(ii) for each $i, 1 \leq i \leq n$,

  a. α_i is a FOL-wff from Γ, or

  b. α_i is of the form $(\omega \vee \sim \omega)$, called *axiom*, where ω is a FOL-wff, or

  c. α_i is a direct consequence of some of the preceding FOL-wffs by virtue of one of the inference rules or by substitution, and

(iii) Each application of **EI** introduces a *new* constant which does not appear in α, and

(iv) No application of **Gen** using x is made to $\alpha(x)$ if the variable x appears as a free variable in a hypothesis upon which $\alpha(x)$ is deduced, and

(v) No application of **Gen** is made using a variable that is free in some $(\exists x)\beta(x)$ to which **EI** has been previously applied.

Conditions (iii), (iv) and (v) are included to ensure that the formal proof approach is sound.

The necessity of condition (iii) can be explained as follows.

Consider the two FOL-wffs $(\exists x)\alpha(x)$ and $(\exists x)\beta(x)$. If we apply EI to both of them, using the same constant k, to obtain $\alpha(k)$ and $\beta(k)$, then we are actually claiming that the element in the universe of discourse satisfying $\alpha(x)$ is the same as that satisfying $\beta(x)$. Clearly, this need not be the case in general. As an example, let $\alpha(x)$: "x is even" and $\beta(x)$: "x is odd." If $\alpha(k)$ is true, then $\beta(k)$ is false, for we cannot have both k is even and k is odd.

Condition (iv) is necessary because any free variable in the hypothesis is treated as a *constant* and as a result cannot be quantified by the universal quantifier. As an example, consider the following statement:

"If x is an even integer greater than 2, then x is not a prime."

The hypothesis of the statement is: "x is an even integer greater than 2." Here, x is not an arbitrary integer but a particular integer having the properties: being even and being greater than 2. Consider the following derivation:

Let $Integer(x)$: x *is an integer*, $Even(x)$: x *is even*, $Divides(x, y)$: x *divides* y, and $Greater(x, y)$: x *is greater than* y.

1.	$Integer(x) \wedge Even(x) \wedge Greater(x, 2)$	hypothesis	
2.	$Integer(x) \wedge Even(x)$	1, I2	
3.	$Integer(x) \wedge Even(x) \Rightarrow Divides(2, x)$		(Definition of even integer)
4.	$Divides(2, x)$	2,3, I3	

5. $(\forall x)Divides(2, x)$ 4, Gen

The FOL-wff in Step 5 asserts that every integer is divisible by 2, which is erroneous. The error arises from the fact that Gen is applied using x in Step 5, where x is free in the hypothesis, thereby violating condition (iv). (Another example is given in Section 2.8, Example 5).

Clearly, if the hypotheses are closed FOL-wffs (i.e., they contain no free variables), then condition (iv) is of no concern to us.

Condition (v) can be explained as follows.

Suppose y is a free variable in $(\exists x)\beta(x)$, and EI has been applied to $(\exists x)\beta(x)$ to obtain $\beta(c)$, where c is a constant. Clearly, y remains as a free variable in $\beta(c)$. Condition (iv) says that Gen cannot be applied to $\beta(c)$ to quantify y. This is because the value of $(\exists x)\beta(x)$ depends on the value of the free variable y. Specifically, for each distinct value of y, the x for which $\beta(x)$ is evaluated to *true* may be different. In other words, the constant c in $\beta(c)$ is a function of y. Therefore, if we apply **Gen** to $\beta(c)$ to quantify y, then the occurrences of c in $\beta(c)$ are 'partially' quantified. This could have undesirable effects (See Example 6 in Section 2.8).

In Chapter 1, we showed that the formal proof approach for Propositional Logic is both sound and complete. Does the same apply to the formal proof approach we have just introduced for First-Order Logic? This is answered affirmatively in the following theorem.

Theorem 2.7.3 (The Soundness and Completeness Theorem of First-order Logic)

A FOL-wff is a theorem iff it is valid

Proof: (Omitted) ■

Therefore, every FOL-wff derivable from a formal proof is a valid FOL-wff, and every valid FOL-wff has a formal proof.

2.7.1 First-order Logic Proof Methods

As with Propositional Logic, the following deduction theorem provides the basis for the ***direct proof*** method in First-Order Logic.

Deduction Theorem: *Let Γ be a set of FOL-wffs and α, β be FOL-wffs. If $\Gamma, \alpha \vdash \beta$, then $\Gamma \vdash \alpha \Rightarrow \beta$.*

Proof: Omitted. ■

All the other proof techniques developed for *propositional logic* can be used for FOL. This is because their correctness is based on the equivalences (E1 — E20) and inference rules (I1 — I6), which remain valid in FOL. In addition, the following variants of Proof by Cases and Bidirectional Proof are useful.

Proof by Cases:

To prove $\vdash (\forall x)(\alpha_1(x) \vee \alpha_2(x) \vee \cdots \vee \alpha_k(x) \Rightarrow \beta(x))$, we may prove $\vdash (\forall x)(\alpha_1(x) \Rightarrow \beta(x))$, $\vdash (\forall x)(\alpha_2(x) \Rightarrow \beta(x))$, . . . , and $\vdash (\forall x)(\alpha_k(x) \Rightarrow \beta(x))$.

Bidirectional Proof:

To prove $\vdash (\forall x)(\alpha(x) \Leftrightarrow \beta(x))$, we may prove $\vdash (\forall x)(\alpha(x) \Rightarrow \beta(x))$ and $\vdash (\forall x)(\beta(x) \Rightarrow \alpha(x))$.

Definition: Let ω be a FOL-wff. To ***disprove*** $\vdash \omega$ is to prove that $\vdash \omega$ does not hold. i.e. ω is *not* a theorem.

The following proof technique is useful in disproving an assertion that a FOL-wff is a theorem.

Proof by Counterexample:

To disprove $\vdash \omega$, it is sufficient to present an interpretation for which ω is evaluated to *false*.

Justification: Let I be an interpretation for which ω is evaluated to *false*.

Then, by definition, ω is invalid.

By the Soundness and Completeness Theorem of FOL, ω is not a theorem. ■

Example 2.7.7 Disprove $\vdash (\exists x)\alpha(x) \wedge (\exists x)\beta(x) \Rightarrow (\exists x)(\alpha(x) \wedge \beta(x))$.

(Proof by Counterexample)

Let the Universe of discourse U be the set of all integers;

$\alpha(x)$: x is an even integer;

$\beta(x)$: x is an odd integer;

As 2 is an even integer and 1 is an odd integer, $(\exists x)\alpha(x)$ and $(\exists x)\beta(x)$ are both *true*.

Since there is no integer that is both even and odd, $(\exists x)(\alpha(x) \wedge \beta(x))$ is *false*.

Hence, $(\exists x)\alpha(x) \wedge (\exists x)\beta(x) \Rightarrow (\exists x)(\alpha(x) \wedge \beta(x))$ is evaluated to false in this interpretation. ■

Example 2.7.8 Disprove $\vdash (\forall x)(\alpha(x) \vee \beta(x)) \Rightarrow (\forall x)\alpha(x) \vee (\forall x)\beta(x)$.

(Proof by Counterexample)

Let the Universe of discourse U be the set of all integers;

$\alpha(x)$: x is an even integer;

$\beta(x)$: x is an odd integer;

As every integer is either even or odd, $(\forall x)(\alpha(x) \vee \beta(x))$ is *true*. But, not every integer is even and not every integer is odd, so both $(\forall x)\alpha(x)$ and $(\forall x)\beta(x)$ are *false*. This implies that $(\forall x)\alpha(x) \vee (\forall x)\beta(x)$ is *false*. Hence, $(\forall x)(\alpha(x) \vee \beta(x)) \Rightarrow (\forall x)\alpha(x) \vee (\forall x)\beta(x)$ is evaluated to *false* in this interpretation. ■

2.8 Examples

In the following examples, $\alpha(x)$ and $\beta(x)$ represent FOL wffs while $P(x), Q(x)$, etc. represent predicates.

1. Prove that $\vdash (\forall x)\alpha(x) \vee (\forall x)\beta(x) \Leftrightarrow (\forall x)(\forall y)(\alpha(x) \vee \beta(y))$ where $\alpha(x)$ and $\beta(x)$ do not contain y.

   (Bidirectional Proof)

   $\Rightarrow$) *(Direct Proof)*

1.	$(\forall x)\alpha(x) \vee (\forall x)\beta(x)$	Hypothesis
2.	$(\forall x)\alpha(x) \vee (\forall y)\beta(y)$	1, FE1, $\beta(x)$ does not contain y
3.	$(\forall x)(\alpha(x) \vee (\forall y)\beta(y))$	2, FE3
4.	$(\forall x)((\forall y)\beta(y) \vee \alpha(x))$	3, E10
5.	$(\forall x)(\forall y)(\beta(y) \vee \alpha(x))$	4, FE3, $\alpha(x)$ does not contain y
6.	$(\forall x)(\forall y)(\alpha(x) \vee \beta(y))$	5, E10

$\Leftarrow$) (*Direct Proof*) List the above derivation in the reversed order.

Hence, $\vdash (\forall x)\alpha(x) \vee (\forall x)\beta(x) \Leftrightarrow (\forall x)(\forall y)(\alpha(x) \vee \beta(y))$ where $\alpha(x)$ and $\beta(x)$ do not contain y. ■

2. Prove that $\vdash (\exists x)\alpha(x) \wedge (\exists x)\beta(x) \Leftrightarrow (\exists x)(\exists y)(\alpha(x) \wedge \beta(y))$ where $\alpha(x)$ and $\beta(x)$ do not contain y.

Solution: A proof similar to that of Example 1 could be easily derived. An alternate proof which makes use of the theorem established in Example 1 is given below.

1. $(\forall x) \sim \alpha(x) \vee (\forall x) \sim \beta(x) \Leftrightarrow (\forall x)(\forall y)(\sim \alpha(x) \vee \sim \beta(y))$ from Γ (The theorem established in Example 1, with $\sim\alpha(x), \sim\beta(x)$ replacing $\alpha(x), \beta(x)$ is used)
2. $\sim((\forall x) \sim \alpha(x) \vee (\forall x) \sim \beta(x)) \Leftrightarrow \sim(\forall x)(\forall y)(\sim \alpha(x) \vee \sim \beta(y))$ 1, Theorem 1.7.2(iii)
3. $\sim(\forall x)\sim\alpha(x) \wedge \sim(\forall x)\sim\beta(x) \Leftrightarrow \sim(\forall x)(\forall y)(\sim \alpha(x) \vee \sim \beta(y))$ 2, E17
4. $(\exists x)\sim\sim\alpha(x) \wedge \sim(\forall x)\sim\beta(x) \Leftrightarrow \sim(\forall x)(\forall y)(\sim \alpha(x) \vee \sim \beta(y))$ 3, FE7
5. $(\exists x)\sim\sim\alpha(x) \wedge (\exists x)\sim\sim\beta(x) \Leftrightarrow \sim(\forall x)(\forall y)(\sim \alpha(x) \vee \sim \beta(y))$ 4, FE7
6. $(\exists x)\sim\sim\alpha(x) \wedge (\exists x)\sim\sim\beta(x) \Leftrightarrow (\exists x)\sim(\forall y)(\sim \alpha(x) \vee \sim \beta(y))$ 5, FE7
7. $(\exists x)\sim\sim\alpha(x) \wedge (\exists x)\sim\sim\beta(x) \Leftrightarrow (\exists x)(\exists y)\sim(\sim\alpha(x) \vee \sim\beta(y))$ 6, FE7
8. $(\exists x)\alpha(x) \wedge (\exists x)\sim\sim\beta(x) \Leftrightarrow (\exists x)(\exists y)\sim(\sim\alpha(x) \vee \sim\beta(y))$ 7, E15
9. $(\exists x)\alpha(x) \wedge (\exists x)\beta(x) \Leftrightarrow (\exists x)(\exists y)\sim(\sim\alpha(x) \vee \sim\beta(y))$ 8, E15
10. $(\exists x)\alpha(x) \wedge (\exists x)\beta(x) \Leftrightarrow (\exists x)(\exists y)(\sim\sim\alpha(x) \wedge \sim\sim\beta(y))$ 9, E17
11. $(\exists x)\alpha(x) \wedge (\exists x)\beta(x) \Leftrightarrow (\exists x)(\exists y)(\alpha(x) \wedge \sim\sim\beta(y))$ 10, E15
12. $(\exists x)\alpha(x) \wedge (\exists x)\beta(x) \Leftrightarrow (\exists x)(\exists y)(\alpha(x) \wedge \beta(y))$ 11, E15

Hence, $\vdash (\exists x)\alpha(x) \wedge (\exists x)\beta(x) \Leftrightarrow (\exists x)(\exists y)(\alpha(x) \wedge \beta(y))$ where $\alpha(x)$ and $\beta(x)$ do not contain y. ■

3. Prove that $\vdash (\exists x)(\alpha(x) \wedge \beta(x)) \Rightarrow (\exists x)\alpha(x) \wedge (\exists x)\beta(x)$.

Solution: (*Direct proof*)

1. $(\exists x)(\alpha(x) \wedge \beta(x))$ Hypothesis
2. $\alpha(c) \wedge \beta(c)$ 1, EI, c is a new constant
3. $\alpha(c)$ 2, I2
4. $\beta(c) \wedge \alpha(c)$ 2, E9
5. $\beta(c)$ 4, I2
6. $(\exists x)\alpha(x)$ 3, EQ
7. $(\exists x)\beta(x)$ 5, EQ
8. $(\exists x)\alpha(x) \wedge (\exists x)\beta(x)$ 6,7, I6

Hence, $\vdash (\exists x)(\alpha(x) \wedge \beta(x)) \Rightarrow (\exists x)\alpha(x) \wedge (\exists x)\beta(x)$ ■

4. Give an example to show that if condition (iii) in the definition of derivation is removed, the resulting formal proof method is not sound.

Solution: Prove that $\vdash (\exists x)(\alpha(x) \Rightarrow \beta(x)) \wedge (\exists x)\alpha(x) \Rightarrow (\exists x)\beta(x)$.

(*Direct Proof*)

1. $(\exists x)(\alpha(x) \Rightarrow \beta(x)) \wedge (\exists x)\alpha(x)$ Hypothesis
2. $(\exists x)(\alpha(x) \Rightarrow \beta(x))$ 1, I2

3. $(\exists x)\alpha(x) \wedge (\exists x)(\alpha(x) \Rightarrow \beta(x))$ 1, E9
4. $(\exists x)\alpha(x)$ 3, I2
5. $\alpha(c) \Rightarrow \beta(c)$ 2, EI, c is a constant
6. $\alpha(c)$ 4, EI, c is a constant
7. $\beta(c)$ 5,6 I3
8. $(\exists x)\beta(x)$ 7, EQ

In Step 5 of the above proof, we apply EI to $(\exists x)\alpha(x)$ using an "old" constant c. This violates condition (iii) in the definition of derivation.

Now, consider evaluating $(\exists x)(\alpha(x) \Rightarrow \beta(x)) \wedge (\exists x)\alpha(x) \Rightarrow (\exists x)\beta(x)$ using the following interpretation. Let the universal of discourse U be $\{a, b\}$ and α, β be defined as follows.

x	$\alpha(x)$
a	T
b	F

x	$\beta(x)$
a	F
b	F

Since $(\exists x)(\alpha(x) \Rightarrow \beta(x)) \wedge (\exists x)\alpha(x)$ is *true* and $(\exists x)\beta(x)$ is *false*, $(\exists x)(\alpha(x) \Rightarrow \beta(x)) \wedge (\exists x)\alpha(x) \Rightarrow (\exists x)\beta(x)$ is evaluated to *false*. Therefore, $(\exists x)(\alpha(x) \Rightarrow \beta(x)) \wedge (\exists x)\alpha(x) \Rightarrow (\exists x)\beta(x)$ is invalid.

While the formal proof establishes the theorem: $\vdash ((\exists x)(\alpha(x) \Rightarrow \beta(x)) \wedge (\exists x)\alpha(x)) \Rightarrow (\exists x)\beta(x)$, the above evaluation shows that $((\exists x)(\alpha(x) \Rightarrow \beta(x)) \wedge (\exists x)\alpha(x)) \Rightarrow (\exists x)\beta(x)$ is invalid. Therefore, the formal proof method is not sound. ■

5. Give an example to show that if condition (iv) in the definition of derivation is removed, the resulting formal proof method is not sound.

Solution: Consider the following formal proof:

Prove that $\vdash E(x) \Rightarrow (\forall x)E(x)$

(Direct Proof)

a. $E(x)$ Hypothesis
b. $(\forall x)E(x)$ 1, Gen

Hence, $\vdash E(x) \Rightarrow (\forall x)E(x)$.

In the above proof, we ignore the fact that x is a free variable in the hypothesis and apply Gen to $E(x)$. This violates condition (iv). Now, if we let the Universe of discourse U to be the set of all positive integers, and

$$E(x) = \begin{cases} true & x \text{ is even and } x > 2 \\ false & otherwise \end{cases}$$

we would have the conclusion: "If x is an even integer greater than 2, then every integer is an even integer greater than 2." which is absurd. Therefore, $E(x) \Rightarrow (\forall x)E(x)$ is evaluated to *false* in this interpretation. It is thus invalid.

Since $\vdash E(x) \Rightarrow (\forall x)E(x)$ and at the same time $E(x) \Rightarrow (\forall x)E(x)$ is invalid, the formal proof method is not sound. ■

6. Give an example to show that if condition (v) in the definition of derivation is removed, the resulting formal proof method is not sound.

Solution: Consider the following formal proof of $\vdash (\forall y)(\exists x)P(x,y) \Rightarrow (\exists x)(\forall y)P(x,y)$
(Direct proof)

1.	$(\forall y)(\exists x)P(x,y)$	Hypothesis
2.	$(\exists x)P(x,y)$	1, UI
3.	$P(c,y)$	2, EI, c is a constant
4.	$(\forall y)P(c,y)$	3, Gen
5.	$(\exists x)(\forall y)P(x,y)$	4, EQ

In Step 4, We apply Gen to quantify the free variable y of $P(c,y)$, where $P(c,y)$ was obtained from an application of EI in Step 3. This violates Condition (v).

Now, consider the following interpretation: let the universe of discourse U be the set of all integers, and $P(x,y)$ be the predicate $Greater(x,y)$ defined earlier. Then $(\forall y)(\exists x)P(x,y)$ would stand for $(\forall y)(\exists x)Greater(x,y)$ and $(\exists x)(\forall y)P(x,y)$ would stand for $(\exists x)(\forall y)Greater(x,y)$. Since "for every integer y, there exists an integer x such that $x>y$" is true (for instance, let x be $y+1$), $(\forall y)(\exists x)Greater(x,y)$ is evaluated to *true*. On the other hand, "there exists an integer x, such that for every integer y, $x>y$" is false because there is no such thing as the largest integer. Therefore, $(\exists x)(\forall y)Greater(x,y)$ is evaluated to *false*. As a result, $(\forall y)(\exists x)P(x,y) \Rightarrow (\exists x)(\forall y)P(x,y)$ is evaluated to *false* in the above interpretation. The formal proof method is thus not sound. ■

7. Prove that:

$\vdash ((\forall x)(P(x) \Rightarrow Q(x)) \wedge (\forall x)(R(x) \Rightarrow S(x))) \Rightarrow (\forall x)((P(x) \wedge R(x)) \Rightarrow (Q(x) \wedge S(x)))$

Solution: (*Direct proof*)

1.	$((\forall x)(P(x) \Rightarrow Q(x)) \wedge (\forall x)(R(x) \Rightarrow S(x)))$	Hypothesis
2.	$(\forall x)((P(x) \Rightarrow Q(x)) \wedge (R(x) \Rightarrow S(x)))$	1, FE9
3.	$(P(x) \Rightarrow Q(x)) \wedge (R(x) \Rightarrow S(x))$	2, UI
4.	$(P(x) \Rightarrow Q(x))$	3, I2
5.	$(R(x) \Rightarrow S(x)) \wedge (P(x) \Rightarrow Q(x))$	3, E9
6.	$(R(x) \Rightarrow S(x))$	5, I2
7.	$\sim P(x) \vee Q(x)$	4, E18
8.	$\sim R(x) \vee S(x)$	6, E18
9.	$(\sim P(x) \vee Q(x)) \vee \sim R(x)$	7, I1
10.	$\sim R(x) \vee (\sim P(x) \vee Q(x))$	9, E10
11.	$(\sim R(x) \vee \sim P(x)) \vee Q(x)$	10, E12
12.	$(\sim P(x) \vee \sim R(x)) \vee Q(x)$	11, E10
13.	$(\sim R(x) \vee S(x)) \vee \sim P(x)$	8, I1
14.	$\sim P(x) \vee (\sim R(x) \vee S(x))$	13, E10
15.	$(\sim P(x) \vee \sim R(x)) \vee S(x)$	14, E12
16.	$((\sim P(x) \vee \sim R(x)) \vee Q(x)) \wedge ((\sim P(x) \vee \sim R(x)) \vee S(x))$	12,15, I6
17.	$(\sim P(x) \vee \sim R(x)) \vee (Q(x) \wedge S(x))$	16, E14
18.	$\sim (P(x) \wedge R(x)) \vee (Q(x) \wedge S(x))$	17, E16
19.	$(P(x) \wedge R(x)) \Rightarrow (Q(x) \wedge S(x))$	18, E18
20.	$(\forall x)((P(x) \wedge R(x)) \Rightarrow (Q(x) \wedge S(x)))$	19, Gen

Hence,

$$\vdash \ ((\forall x)(P(x) \Rightarrow Q(x)) \wedge (\forall x)(R(x) \Rightarrow S(x))) \ \Rightarrow \ (\forall x)((P(x) \wedge R(x)) \Rightarrow (Q(x) \wedge S(x)))$$

■

8. Given P1: $(\forall x)(P(x) \Rightarrow Q(x))$

P2: $(\exists x)(R(x) \wedge (\forall y)(Q(y) \Rightarrow \sim S(x,y)))$

C: $(\exists x)(R(x) \wedge (\forall y)(P(y) \Rightarrow \sim S(x,y)))$

Prove that P1, P2 $\vdash$ C.

Solution:

1.	$(\forall x)(P(x) \Rightarrow Q(x))$	from Γ
2.	$(\exists x)(R(x) \wedge (\forall y)(Q(y) \Rightarrow \sim S(x,y)))$	from Γ
3.	$R(k) \wedge (\forall y)(Q(y) \Rightarrow \sim S(k,y))$	2, EI, k is a (new) constant
4.	$P(y) \Rightarrow Q(y)$	1, UI
5.	$(\forall y)(Q(y) \Rightarrow \sim S(k,y)) \wedge R(k)$	3, E9
6.	$(\forall y)(Q(y) \Rightarrow \sim S(k,y))$	5, I2
7.	$Q(y) \Rightarrow \sim S(k,y)$	6, UI
8.	$P(y) \Rightarrow \sim S(k,y)$	4,7, I5
9.	$(\forall y)(P(y) \Rightarrow \sim S(k,y))$	8, Gen
10.	$R(k)$	3, I2
11.	$R(k) \wedge (\forall y)(P(y) \Rightarrow \sim S(k,y))$	10,9, I6
12.	$(\exists x)(R(x) \wedge (\forall y)(P(y) \Rightarrow \sim S(x,y)))$	11, EQ

Hence, P1, P2 $\vdash$ C. ■

9. Given P1: $(\forall x)(\exists y)(P(x,y) \wedge S(x,y))$

P2: $(\forall x)(\forall y)(P(x,y) \Rightarrow R(x,y))$

C: $(\forall x)(\exists y)(R(x,y) \wedge S(x,y))$

Prove that P1, P2 $\vdash$ C.

Solution:

1.	$(\forall x)(\exists y)(P(x,y) \wedge S(x,y))$	from Γ
2.	$(\forall x)(\forall y)(P(x,y) \Rightarrow R(x,y))$	from Γ
3.	$(\exists y)(P(x,y) \wedge S(x,y))$	1, UI
4.	$P(x,k) \wedge S(x,k)$	3, EI, k is a (new) constant
5.	$(\forall y)(P(x,y) \Rightarrow R(x,y))$	2, UI
6.	$P(x,k) \Rightarrow R(x,k)$	5, UI
7.	$P(x,k)$	4, I2
8.	$S(x,k) \wedge P(x,k)$	4, E9
9.	$S(x,k)$	8, I2
10.	$R(x,k)$	7,6, I3
11.	$R(x,k) \wedge S(x,k)$	10,9, I6
12.	$(\exists y)(R(x,y) \wedge S(x,y))$	11, EQ
13.	$(\forall x)(\exists y)(R(x,y) \wedge S(x,y))$	2, Gen

Hence, P1, P2 $\vdash$ C. ■

10. Given P1 : $(\forall x)(P(x) \Rightarrow (\exists y)(Q(y) \wedge S(x,y)))$
P2 : $\sim (\exists x)(\exists y)(P(x) \wedge R(y) \wedge S(x,y))$
C : $(\exists x)P(x) \Rightarrow (\exists y)(Q(y) \wedge \sim R(y))$
Prove that P1, P2 $\vdash$ C.

Solution: *(Proof by contradiction)*

1.	$\sim ((\exists x)P(x) \Rightarrow (\exists y)(Q(y) \wedge \sim R(y)))$	Hypothesis
2.	$(\forall x)(P(x) \Rightarrow (\exists y)(Q(y) \wedge S(x,y)))$	from Γ
3.	$\sim (\exists x)(\exists y)(P(x) \wedge R(y) \wedge S(x,y))$	from Γ
4.	$\sim (\sim (\exists x)P(x) \vee (\exists y)(Q(y) \wedge \sim R(y)))$	1, E18
5.	$\sim\sim (\exists x)P(x) \wedge \sim (\exists y)(Q(y) \wedge \sim R(y))$	4, E17
6.	$(\exists x)P(x) \wedge \sim (\exists y)(Q(y) \wedge \sim R(y))$	5, E15
7.	$(\exists x)P(x) \wedge (\forall y) \sim (Q(y) \wedge \sim R(y))$	6, FE8
8.	$(\exists x)P(x)$	7, I2
9.	$(\forall y) \sim (Q(y) \wedge \sim R(y)) \wedge (\exists x)P(x)$	7, E9
10.	$(\forall y) \sim (Q(y) \wedge \sim R(y))$	9, I2
11.	$(\forall y)(\sim Q(y) \vee \sim\sim R(y))$	10, E16
12.	$(\forall y)(\sim Q(y) \vee R(y))$	11, E15
13.	$(\forall x) \sim (\exists y)(P(x) \wedge R(y) \wedge S(x,y))$	3, FE8
14.	$(\forall x)(\forall y) \sim (P(x) \wedge R(y) \wedge S(x,y))$	13, FE8
15.	$(\forall x)(\forall y)(\sim (P(x) \wedge R(y)) \vee \sim S(x,y))$	14, E16
16.	$(\forall x)(\forall y)((\sim P(x) \vee \sim R(y)) \vee \sim S(x,y))$	15, E16
17.	$P(k)$	8, EI, k is a (new) constant
18.	$P(k) \Rightarrow (\exists y)(Q(y) \wedge S(k,y))$	2, UI
19.	$(\exists y)(Q(y) \wedge S(k,y))$	17,18, I3
20.	$Q(b) \wedge S(k,b)$	19, EI, b is a (new) constant
21.	$\sim Q(b) \vee R(b)$	12, UI
22.	$Q(b)$	20, I2
23.	$Q(b) \Rightarrow R(b)$	21, E18
24.	$R(b)$	22,23, I3
25.	$(\forall y)((\sim P(k) \vee \sim R(y)) \vee \sim S(k,y))$	16, UI
26.	$(\sim P(k) \vee \sim R(b)) \vee \sim S(k,b)$	25, UI
27.	$(\sim R(b) \vee \sim P(k)) \vee \sim S(k,b)$	26, E10
28.	$\sim R(b) \vee (\sim P(k) \vee \sim S(k,b))$	27, E12
29.	$R(b) \Rightarrow (\sim P(k) \vee \sim S(k,b))$	28, E18
30.	$\sim P(k) \vee \sim S(k,b)$	24,29, I3
31.	$P(k) \Rightarrow \sim S(k,b)$	30, E18
32.	$\sim S(k,b)$	17,31, I3
33.	$S(k,b) \wedge Q(b)$	20, E9
34.	$S(k,b)$	33, I2
35.	$S(k,b) \wedge \sim S(k,b)$	34,32, I6
36.	$false$	35, E1

Hence, P1, P2 $\vdash$ C ■

2.9 Applications of FOL

As with Propositional Logic, First-Order Logic can be used to prove the validity of a valid argument.

First, we shall discuss how to translate an English sentence into a FOL-wff with a number of examples.

Example 1. Every bird has two wings.

Solution:

Let U(the Universe of discourse) be the set of all the things;

$B(x)$ denote x is a bird;

$W(x)$ denote x has two wings

Then the corresponding FOL-wff is: $(\forall x)(B(x) \Rightarrow W(x))$ ■

Remark:

(i) The following translation is incorrect: $(\forall x)(B(x) \wedge W(x))$. This is because it corresponds to the sentence: "*Everything is a bird which has two wings*" which is different from the given sentence.

(ii) In general, statement beginning with "Every" or "All" are translated into $(\forall x)(\alpha(x) \Rightarrow \beta(x))$.

(iii) If U is defined as the set of all birds, then the corresponding FOL-wff could be $(\forall x)W(x)$.

Example 2. All the babies are cute.

Solution:

Let U(the Universe of discourse) be the set of people;

$B(x)$ denote x is a baby;

$C(x)$ denote x is cute

Then the corresponding FOL-wff is: $(\forall x)(B(x) \Rightarrow C(x))$ ■

Example 3. An illogical person is a irrational person.

Solution:

Let U(the Universe of discourse) be the set of people;

$I(x)$ denote x is illogical;

$R(x)$ denote x is rational

Then the corresponding FOL-wff is: $(\forall x)(I(x) \Rightarrow \sim R(x))$ ■

Example 4. Some students are lazy.

Solution:

Let U(the Universe of discourse) be the set of people;

$S(x)$ denote x is a student;

$L(x)$ denote x is lazy

Then the corresponding FOL-wff is: $(\exists x)(S(x) \wedge L(x))$ ■

Remark:

(i) The following translation is incorrect: $(\exists x)(S(x) \Rightarrow L(x))$. This is because $(\exists x)(S(x) \Rightarrow L(x))$ is equivalent to $(\exists x)(\sim S(x) \vee L(x))$ which asserts that there is a person who is either not a student or is lazy which is quite different from *some students are lazy*.

(ii) In general, statement beginning with "Some" or "There is at least one" are translated into $(\exists x)(\alpha(x) \wedge \beta(x))$.

Example 5. In every city, there is at least a postman who has been bitten by every dog in that city.

Solution:

Let U(the Universe of discourse) be the set of all things;

$C(x)$ denote x is a city;
$P(x)$ denote x is a postman;
$D(x)$ denote x is a dog;
$B(x,y)$ denote x has been bitten by y.
$L(x,y)$ denote x lives in y.

Then the corresponding FOL-wff is:

$(\forall w)(C(w) \Rightarrow (\exists x)(P(x) \wedge L(x,w) \wedge (\forall y)(D(y) \wedge L(y,w) \Rightarrow B(x,y))))$ ■

Example 6. Everyone lies to his friend in sometime of his life.

Solution:

Let U(the Universe of discourse) be the set of all conceivable concepts (objects);

$P(x)$ denote x is a person;
$T(x)$ denote x is time;
$F(x,y)$ denote x is a friend of y;
$L(x,y,t)$ denote x lies to y at time t.

Then the corresponding FOL-wff is:

$(\forall x)(P(x) \Rightarrow (\exists y)(P(y) \wedge F(x,y) \wedge (\exists t)(T(t) \wedge L(x,y,t))))$ ■

Example 7. No used-car dealer buys a used car for his family.

Some people who buy used cars for their families are absolutely dishonest.

Conclusion: Some absolutely dishonest people are not used-car dealers.

Solution: Let U(the Universe of discourse) be the set of people;

$U(x)$ denote x is a used-car dealer;
$B(x)$ denote x buys a used car for his family;
$D(x)$ denote x is dishonest;

Then, the corresponding FOL-wffs are as follows.

P1: $\sim (\exists x)(U(x) \wedge B(x))$

P2: $(\exists x)(B(x) \wedge D(x))$

C: $(\exists x)(D(x) \wedge \sim U(x))$

We shall prove that P1, P2 $\vdash$ C.

1.	$\sim (\exists x)(U(x) \wedge B(x))$	from Γ
2.	$(\exists x)(B(x) \wedge D(x))$	from Γ
3.	$B(k) \wedge D(k)$	2, EI, k is a constant
4.	$(\forall x) \sim (U(x) \wedge B(x))$	1, FE8
5.	$\sim (U(k) \wedge B(k))$	4, UI
6.	$\sim U(k) \vee \sim B(k)$	5, E16
7.	$U(k) \Rightarrow \sim B(k)$	6, E18
8.	$B(k)$	3, I2
9.	$\sim\sim B(k)$	8, E15
10.	$\sim U(k)$	7,9, I4
11.	$D(k) \wedge B(k)$	3, E9
12.	$D(k)$	11, I2
13.	$D(k) \wedge \sim U(k)$	12,10, I6
14.	$(\exists x)(D(x) \wedge \sim U(x))$	13, EQ

Hence, P1, P2 $\vdash$ C ■

Example 8 Some patients like all doctors.

No patient likes any quack.

Therefore, no doctor is a quack.

Solution: Let U(the Universe of discourse) be the set of people;

$P(x)$ denote x is a patient;
$D(x)$ denote x is a doctor;
$Q(x)$ denote x is a quack;
$L(x,y)$ denote x likes y;

Then, the corresponding FOL-wffs are as follows.

P1: $(\exists x)(P(x) \wedge (\forall y)(D(y) \Rightarrow L(x, y)))$

P2: $\sim (\exists x)(\exists y)(P(x) \wedge Q(y) \wedge L(x, y))$

C: $\sim (\exists x)(D(x) \wedge Q(x))$

We shall prove that: P1, P2 $\vdash$ C.

1.	$(\exists x)(P(x) \wedge (\forall y)(D(y) \Rightarrow L(x, y)))$	from Γ
2.	$\sim (\exists x)(\exists y)(P(x) \wedge Q(y) \wedge L(x, y))$	from Γ
3.	$(P(k) \wedge (\forall y)(D(y) \Rightarrow L(k, y)))$	1, EI, k is a new constant
4.	$P(k)$	3, I2
5.	$(\forall y)(D(y) \Rightarrow L(k, y)) \wedge P(k)$	3, E9
6.	$(\forall y)(D(y) \Rightarrow L(k, y))$	5, I2
7.	$D(y) \Rightarrow L(k, y)$	6, UI
8.	$(\forall x) \sim (\exists y)(P(x) \wedge Q(y) \wedge L(x, y))$	2, FE8
9.	$(\forall x)(\forall y) \sim (P(x) \wedge Q(y) \wedge L(x, y))$	8, FE8
10.	$(\forall y) \sim (P(k) \wedge Q(y) \wedge L(k, y))$	9, UI
11.	$\sim (P(k) \wedge Q(y) \wedge L(k, y))$	10, UI
12.	$\sim (P(k) \wedge (Q(y) \wedge L(k, y)))$	11, E11
13.	$\sim P(k) \vee \sim (Q(y) \wedge L(k, y))$	12, E16